

JAYDEN KIM

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Engineering, Minor in Statistics

Expected Dec 2026

- **Coursework:** Operating Systems, Algorithms & Models of Computation, Distributed Systems, Data Structures, Machine Learning Systems, Applied Machine Learning, Deep Learning for Computer Vision, Database Systems
- **Awards/Honors:** USACO Gold Division, Holonyak Scholarship (\$5,000)

TECHNICAL SKILLS

Languages:	Java, Python, JavaScript, TypeScript, SQL, C, C++, HTML/CSS, RISC-V Assembly
Frameworks & Libraries:	React, Node.js, FastAPI, PyTorch, Scikit-learn, Vitest, Playwright
Databases & Tools:	PostgreSQL, MongoDB, AWS, GCP, Docker, Git, GitHub, Linux, Bash, GDB, Stripe API, Modal
Practices:	SDLC, Web Services, REST APIs, API Integration, CI/CD, Unit & E2E Testing, Code Reviews

EXPERIENCE

Merkaloop

Jun 2026 – Present

Software Engineering Intern

Remote

- Translated business requirements into a rules-based **TypeScript** pricing engine that values used lab equipment from retail-price anchors, depreciation curves, and condition modifiers, returning price ranges with confidence scores.
- Debugged a currency-conversion defect in the calibration pipeline that silently discarded all EUR-priced listings, and added versioned, append-only **database** tables so every valuation is reproducible and auditable.
- Fixed an **e-commerce checkout** race condition in the **Stripe** payment flow that could sell one unit to two buyers, replacing a read-then-write with an atomic **PostgreSQL** update to prevent overselling.
- Built an LLM agent that calls live web search to find comparable competitor listings for **89%** of catalog items and surfaces the lowest price to guide seller pricing.

Printech Workshop

Apr 2026 – Jun 2026

Software Engineering Intern

Chicago, IL

- Built a full-stack ERP web application with **Python (FastAPI)** and **Node.js**, integrating **Wix** and **Discord REST APIs** to unify order, inventory, and production data for **250+** products and save **~10 hrs/week** of manual work.
- Designed a **demand forecasting** model on historical sales data that ranks products by projected demand to set the production queue, cutting top-seller **inventory stockouts** from **25** to **2** per month.
- Wrote a job scheduler that groups print jobs sharing the same machine settings, reducing press recalibrations from **9** to **3** per shift and recovering **80 min/day** of downtime.

US Navy

May 2024 – Aug 2024

Software Engineering Intern (Contract)

Lakehurst, NJ

- Developed a **Python/PostgreSQL** business system that automates support equipment tracking for VH-92 helicopter operations, eliminating **80%** of manual data entry.
- Containerized **ETL pipelines** with **Docker** and set up **CI/CD** workflows for automated build, test, and deployment, cutting release time by **50%**.

PROJECTS

AlignAI | [GitHub](#) | [Demo](#)

Feb 2026 – Mar 2026

- Built the pose-comparison engine in **Python** for a 3-person hackathon fitness app, aligning user and reference workouts with dynamic time warping over normalized joint angles, served as a **REST API** on **Modal**.
- Reduced search latency with a prefix-aware TTL cache in front of a third-party exercise-search API, plus client-side debouncing that cancels stale requests with **AbortController**.
- Set up the testing stack for the **React/TypeScript** app, writing **Vitest** unit tests and **Playwright** end-to-end tests.

Operating System Kernel from Scratch | [GitHub](#) | [Demo](#)

Aug 2025 – Dec 2025

- Engineered a Unix-like kernel from scratch in **C** and **RISC-V assembly** with preemptive multitasking, Sv39 paging, and a read-write filesystem (KTFS) with write-back cache.
- Implemented **system calls** (fork, exec, wait) and a custom shell supporting **I/O redirection**, **piping**, and background jobs.
- Wrote **VirtIO** block and UART drivers plus an **ELF loader** running user binaries in isolated virtual memory, booting to an interactive shell on **QEMU**.